



# SAFETY DATA SHEET

Revision Date 31-Mar-2023

WAI1 - AGHS - OSHA

Revision Number 3

## 1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING

### Product Identifier

**Product Name** Oxygen Scavenger Reagent

**Product No** 181811

**Pure substance/mixture** Mixture

Contains Acetic Acid, Iodine

### Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Use as laboratory reagent

**Uses advised against** No Information available

**Manufacturer, Importer, Supplier** Thermo Fisher Scientific©  
Water and Lab Products  
22 Alpha Road  
Chelmsford, MA 01824, USA  
1-978-232-6000

**Emergency Telephone** 24 Hour Emergency Phone Number  
CHEMTREC®  
Within USA and Canada: 1-800-424-9300  
Outside USA and Canada: 1-703-527-3887  
(collect calls accepted)

**E-mail address** [info.water@thermo.com](mailto:info.water@thermo.com)

**Made in** USA

## 2. HAZARDS IDENTIFICATION

### Classification

#### **OSHA Regulatory Status**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                      |            |
|------------------------------------------------------|------------|
| Skin Corrosion/Irritation                            | Category 1 |
| Serious Eye Damage/Eye Irritation                    | Category 1 |
| Specific target organ toxicity - (repeated exposure) | Category 1 |

### Label Elements

Contains Acetic Acid, Iodine

### Emergency Overview

#### **Danger**

#### **Hazard Statements**

Causes severe skin burns and eye damage  
Causes damage to organs through prolonged or repeated exposure  
Toxic to aquatic life



**Appearance** Dark red

**Physical State** Liquid

**Odor** vinegar-like

### Precautionary Statements

#### **Prevention**

Wear protective gloves/protective clothing/eye protection/face protection  
Do not breathe dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product

#### **Response**

##### **Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER or doctor/physician

##### **Skin**

Call a POISON CENTER or doctor/physician if you feel unwell  
Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

##### **Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Immediately call a POISON CENTER or doctor/physician

##### **Ingestion**

IF SWALLOWED: Rinse mouth. DO NOT induce vomiting

#### **Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

No information available

**Other Information**

Toxic to aquatic organisms

### 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component   | CAS No    | Weight % |
|-------------|-----------|----------|
| Acetic Acid | 64-19-7   | 60 - 70% |
| Water       | 7732-18-5 | 10 - 20% |
| Iodine      | 7553-56-2 | 10 - 20% |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. FIRST AID MEASURES

**Description of first aid measures**

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                        |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.                                                                                                                                                                                           |
| <b>Self-Protection of the First Aider</b> | Use personal protective equipment as required. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.                           |

**Most important symptoms and effects, both acute and delayed**

|                                            |                                                             |
|--------------------------------------------|-------------------------------------------------------------|
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes Difficulty in breathing |
|--------------------------------------------|-------------------------------------------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                           |                       |
|---------------------------|-----------------------|
| <b>Notes to Physician</b> | Treat symptomatically |
|---------------------------|-----------------------|

### 5. FIRE-FIGHTING MEASURES

**Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Unsuitable Extinguishing Media**

No information available

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Explosion Data**

**Sensitivity to Mechanical Impact** None

**Sensitivity to Static Discharge** None

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## 6. ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

**Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system. Vapors may accumulate to form explosive concentrations.

**Methods and Material for Containment and Cleaning Up**

**Methods for Containment**

Prevent further leakage or spillage if safe to do so.

**Methods for Cleaning Up**

Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

## 7. HANDLING AND STORAGE

**Precautions for Safe Handling**

**Handling**

Wear personal protective equipment/face protection  
Do not get in eyes, on skin, or on clothing  
Use only under a chemical fume hood  
Do not breathe mist/vapors/spray  
Do not ingest. If swallowed then seek immediate medical assistance  
Keep away from open flames, hot surfaces and sources of ignition  
Use only non-sparking tools  
Take precautionary measures against static discharges

**Conditions for Safe Storage, Including any Incompatibilities**

**Storage**

Corrosives area  
Keep containers tightly closed in a dry, cool and well-ventilated place  
Keep away from heat, sparks and flame

**Incompatible Products**

No information available

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control parameters**

## Exposure Guidelines

| Component              | ACGIH TLV                      | OSHA PEL                                                                                                                 | NIOSH                                                                                                  |
|------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Acetic Acid<br>64-19-7 | TWA: 10 ppm<br>STEL: 15 ppm    | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 25 mg/m <sup>3</sup><br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                 | IDLH: 50 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup> |
| Iodine<br>7553-56-2    | TWA: 0.01 ppm<br>STEL: 0.1 ppm | Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup><br>(Vacated) Ceiling: 0.1 ppm<br>(Vacated) Ceiling: 1 mg/m <sup>3</sup> | IDLH: 2 ppm<br>Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup>                                        |

## Legend

ACGIH TLV: American Conference of Governmental Industrial Hygienists - Threshold Limit Value

OSHA PEL: Occupational Safety and Health Administration - Permissible Exposure Limits

NIOSH - National Institute for Occupational Safety and Health IDLH - Immediately Dangerous to Life or Health

## Appropriate engineering controls

**Engineering Measures** Ensure that eyewash stations and safety showers are close to the workstation location  
Ensure adequate ventilation, especially in confined areas Use explosion-proof electrical/ventilating/lighting equipment

## Individual protection measures, such as personal protective equipment

**Eye/face Protection** Wear chemical splash goggles and face shield. If splashes are likely to occur: Face protection shield.

**Skin and Body Protection** Wear protective gloves/protective clothing.

**Respiratory Protection** None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical State | Liquid                   |
| Appearance     | Dark red                 |
| Odor           | vinegar-like             |
| Odor Threshold | No information available |
| pH             | 1.8                      |
| PH Range       | 0.3 - 3.3                |

| Property                     | Values                   | Remarks • Method |
|------------------------------|--------------------------|------------------|
| Melting point/freezing point | No information available |                  |
| Boiling Point/Range          | ~ 100 °C / 212 °F        |                  |
| Flash Point (High in °C)     | > N/A °C / > 141.8 °F    |                  |
| Evaporation Rate             | No information available |                  |
| Flammability (solid, gas)    | No information available |                  |
| Flammability Limit in Air    |                          |                  |
| Upper flammability limit:    | No information available |                  |
| Lower flammability limit:    | No information available |                  |
| Vapor pressure               | No information available |                  |
| Vapor Density                | No information available |                  |

|                                     |                          |
|-------------------------------------|--------------------------|
| <b>Specific Gravity</b>             | No information available |
| <b>Water Solubility</b>             | Soluble in water         |
| <b>Solubility in other solvents</b> | No information available |
| <b>Partition coefficient</b>        | No information available |
| <b>Autoignition Temperature</b>     | -                        |
| <b>Decomposition Temperature</b>    | No information available |
| <b>Kinematic viscosity</b>          | No information available |
| <b>Dynamic viscosity</b>            | No information available |
| <b>Explosive Properties</b>         |                          |
| <b>Oxidizing Properties</b>         | No information available |

**Other Information**

|                         |                          |
|-------------------------|--------------------------|
| <b>Softening Point</b>  | No information available |
| <b>Molecular Weight</b> | No information available |
| <b>VOC Content(%)</b>   | 61.2                     |
| <b>Density</b>          | No Information available |
| <b>Bulk Density</b>     | No information available |

**10. STABILITY AND REACTIVITY**

**Reactivity**

No Information available

**Chemical Stability**

Stable under normal conditions

**Possibility of Hazardous Reactions**

None under normal processing

**Conditions to Avoid**

Keep away from open flames, hot surfaces and sources of ignition.

**Incompatible Materials**

No information available

**Hazardous Decomposition Products**

Thermal decomposition can lead to release of irritating gases and vapors.

**11. TOXICOLOGICAL INFORMATION**

**Information on likely routes of exposure**

**Product Information**

|                     |                                                                       |
|---------------------|-----------------------------------------------------------------------|
| <b>Inhalation</b>   | No information available                                              |
| <b>Eye Contact</b>  | Corrosive to the eyes and may cause severe damage including blindness |
| <b>Skin Contact</b> | Causes burns                                                          |
| <b>Ingestion</b>    | No information available                                              |

| Component              | LD50 Oral                 | LD50 Dermal                                                  | LC50 Inhalation               |
|------------------------|---------------------------|--------------------------------------------------------------|-------------------------------|
| Acetic Acid<br>64-19-7 | LD50 = 3310 mg/kg ( Rat ) | LD50 = 1060 mg/kg ( Rabbit )                                 | LC50 = 11.4 mg/L ( Rat ) 4 h  |
| Water<br>7732-18-5     | LD50 > 90 mL/kg ( Rat )   | -                                                            | -                             |
| Iodine<br>7553-56-2    | LD50 = 14 g/kg ( Rat )    | LD50 > 2000 mg/kg ( Rabbit )<br>LD50 = 1425 mg/kg ( Rabbit ) | LC50 > 4.588 mg/L ( Rat ) 4 h |

### Information on Toxicological Effects

**Symptoms** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting  
Product is a corrosive material. Use of gastric lavage or emesis is contraindicated.  
Possible perforation of stomach or esophagus should be investigated Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Sensitization** No information available

**Mutagenic Effects** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component   | CAS No    | IARC       | NTP        | ACGIH      | OSHA       |
|-------------|-----------|------------|------------|------------|------------|
| Acetic Acid | 64-19-7   | Not listed | Not listed | Not listed | Not listed |
| Water       | 7732-18-5 | Not listed | Not listed | Not listed | Not listed |
| Iodine      | 7553-56-2 | Not listed | Not listed | Not listed | Not listed |

**Reproductive Effects** No information available

**STOT - single exposure** No information available

**STOT - repeated exposure** Causes damage to organs through prolonged or repeated exposure

**Aspiration hazard** No information available

### Numerical measures of toxicity - Product Information

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral) 5031 mg/kg  
ATEmix (dermal) 1327 mg/kg  
ATEmix (inhalation-dust/mist) 5.5 mg/L

## **12. ECOLOGICAL INFORMATION**

### Ecotoxicity

0% of the mixture consists of components(s) of unknown hazards to the aquatic environment

| Component              | Freshwater Algae | Freshwater Fish                                                                                              | Water Flea                                     |
|------------------------|------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Acetic Acid<br>64-19-7 | -                | LC50: = 75 mg/L, 96h static<br>(Lepomis macrochirus)<br>LC50: = 79 mg/L, 96h static<br>(Pimephales promelas) | EC50: = 65 mg/L, 48h Static<br>(Daphnia magna) |
| Iodine<br>7553-56-2    | -                | LC50: = 1.67 mg/L, 96h static<br>(Oncorhynchus mykiss)                                                       | -                                              |

### Persistence and Degradability

No information available

### Bioaccumulation/ Accumulation

No information available

### Mobility

| Component              | log Pow |
|------------------------|---------|
| Acetic Acid<br>64-19-7 | -0.17   |

**Other adverse effects**  
No information available

### 13. DISPOSAL CONSIDERATIONS

#### Waste treatment methods

**Waste Disposal Methods** Disposal should be in accordance with applicable regional, national and local laws and regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

| Component              | CAWAST                          |
|------------------------|---------------------------------|
| Acetic Acid<br>64-19-7 | Toxic<br>Corrosive<br>Ignitable |

### 14. TRANSPORT INFORMATION

#### DOT

UN-No UN2790  
Proper Shipping Name ACETIC ACID SOLUTION  
Hazard Class 8  
Packing Group II  
Reportable Quantity (RQ): (Acetic Acid: RQ (kg)= 2270.00)  
Special Provisions 148, A3, A7, A10, B2, IB2, T7, TP2  
Shipping Description UN2790, ACETIC ACID SOLUTION, 8, II  
Emergency Response Guide Number 153

#### ICAO

UN-No UN2790  
Proper Shipping Name ACETIC ACID SOLUTION  
Hazard Class 8  
Packing Group II  
Description UN2790, ACETIC ACID SOLUTION, 8, II

#### IATA

UN-No UN2790  
Proper Shipping Name ACETIC ACID SOLUTION  
Hazard Class 8  
Packing Group II  
ERG Code 8L  
Description UN2790, ACETIC ACID SOLUTION, 8, II

#### IMDG/IMO

UN-No UN2790  
Proper Shipping Name ACETIC ACID SOLUTION  
Hazard Class 8  
Packing Group II  
EmS No. F-A, S-B  
Description UN2790, ACETIC ACID SOLUTION, 8, II, Marine Pollutant

### 15. REGULATORY INFORMATION



**International Inventories**

|                                           |          |
|-------------------------------------------|----------|
| <b>United States of America Inventory</b> | Complies |
| <b>CANINV</b>                             | Complies |
| <b>EINECS/ELINCS</b>                      | Complies |
| <b>ENCS</b>                               | Complies |
| <b>IECSC</b>                              | Complies |
| <b>KECL</b>                               | Complies |
| <b>PICCS</b>                              | Complies |
| <b>AICS</b>                               | Complies |

**Legend:**

**USINV/ TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**CANINV/ DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances

**U.S. Federal Regulations**

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications. Under the amended regulations at 40 CFR 370, EPCRA 311/312 Tier II reporting for the 2017 calendar year will need to be consistent with updated hazard classifications.

**CWA (Clean Water Act)**

| Component              | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants | CWA - Hazardous Substances |
|------------------------|-----------------------------|------------------------|---------------------------|----------------------------|
| Acetic Acid<br>64-19-7 | 5000 lb                     | -                      | -                         | X                          |

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

| Component              | Hazardous Substances RQs | CERCLA EHS RQs | RQ                                         |
|------------------------|--------------------------|----------------|--------------------------------------------|
| Acetic Acid<br>64-19-7 | 5000 lb                  | -              | RQ 5000 lb final RQ<br>RQ 2270 kg final RQ |

**U.S. State Regulations**

**California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Component              | New Jersey | Massachusetts | Pennsylvania |
|------------------------|------------|---------------|--------------|
| Acetic Acid<br>64-19-7 | X          | X             | X            |
| Water<br>7732-18-5     | -          | -             | X            |
| Iodine<br>7553-56-2    | X          | X             | X            |

**U.S. EPA Label Information**

No information available

**16. OTHER INFORMATION**

|                            |                                |
|----------------------------|--------------------------------|
| <b>Prepared By</b>         | Regulatory Affairs             |
| <b>Prepared For</b>        | Thermo Fisher Scientific Inc.© |
| <b>Issue Date</b>          | No information available       |
| <b>Revision Date</b>       | 31-Mar-2023                    |
| <b>Reason for revision</b> | SDS sections updated.          |

**Disclaimer**

IMPORTANT: The information contained in this SDS is correct to the best of our knowledge as of the issue date (or subsequent revision date, if any), and is to be used only as a guide. This SDS does not constitute a guarantee (express or implied) of any kind and we make no warranties of any kind as to the accuracy or completeness of the information contained herein or the merchantability or fitness of the product or this information for a particular purpose. It is the responsibility of each individual buyer/user to determine the suitability of this information and the product for its intended purposes. Product sales are subject to Thermo Fisher Scientifics standard terms and conditions of sale. This information relates only to the designated product as shipped and may not be valid if the product is used in combination with any other materials or is not used in accordance with our instructions, or is altered in any way. It is the responsibility of the buyer/user to ensure that its activities comply with all applicable government requirements. Since conditions of use of the product are not under direct control of Thermo Fisher Scientific, it is the duty of the buyer/user to determine the necessary conditions for the safe use of the product. Thermo Fisher Scientific will not be liable for any injuries or damages resulting from handling, use, misuse or contact with the product.

**End of Safety Data Sheet**